

# TNC / Rideshare Insurance Platform

Specialty commercial auto insurance with tiered coverage derived from driver app status at the instant of impact.

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ARCHITECTURE BASELINE

.NET 8 • Angular 21 • Clean Architecture

# TNC Coverage Disputes & Adjudication Strategy

## ⚠ The Industry Challenge

Under statutes like Texas Insurance Code Ch. 1954, coverage varies by driver app state. State disputes at impact cause major claim friction.

**Period 1:** App on, awaiting ride request

**Period 2:** Match accepted, en route to pickup

**Period 3:** Passenger on board until drop-off

## 🛡 Our Architectural Solution

Deterministic automation replaces manual assertions through timestamped trip event derivation and straight-through binding.

**Deterministic Derivation:** Period matched against trip logs

**Straight-Through Binding:** Standard pricing acts as underwriter

**Section 64UM Adherence:** Surveyor field reviews where funds leave

# Verified Functional Scope (FR1 – FR9)

**FR1: Auth & Roles**

In-memory JWT + HttpOnly refresh cookie;  
Company/Surveyor/Admin.

**FR2: Company Profile**

Single profile with CIN validation; 403 on foreign  
profile access.

**FR2A: Surveyor Vetting**

IRDAI licence capture; 403 lock until Admin  
approval granted.

**FR3: Quote Management**

Tier A/B quotes; live FX rates (USD/EUR); 7-day  
lazy expiration.

**FR4: Policy Engine**

Straight-through binding; 365-day term; claims-  
ratio renewal bands.

**FR5: Claims Adjudication**

Derived period rules; 4-field survey reviews;  
mandatory override notes.

**FR6: Simulated Payments**

Deterministic premium activation and claim  
payout settlement.

**FR7: Admin Dashboard**

5 KPIs: Policies, Claims, Payouts, Approval Ratio,  
Conversion Rate.

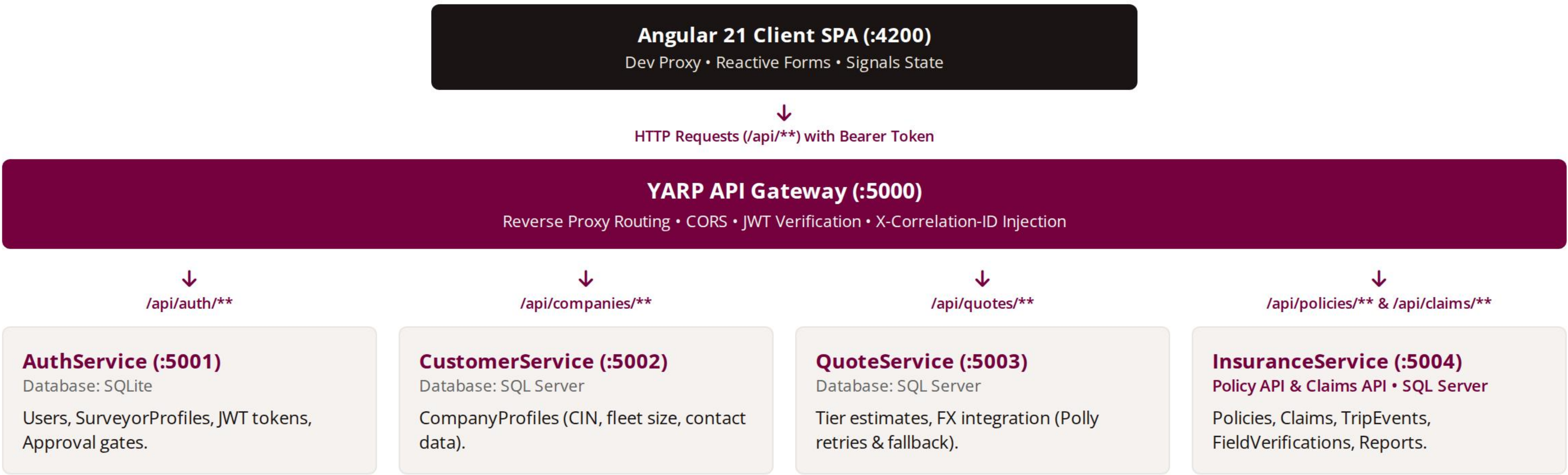
**FR8/9: Resilience & Paging**

Polly FX retry; health probes; correlation IDs;  
server-side paging.

# Complete Technology Stack

| LAYER          | TECHNOLOGY & IMPLEMENTATION DETAILS                                            |
|----------------|--------------------------------------------------------------------------------|
| UI             | Angular 21 (Standalone Components, Signals, Reactive Forms, RxJS)              |
| Styling        | Tailwind CSS 4 (The Hartford Brand Design Tokens declared via @theme)          |
| API            | ASP.NET Core 8 Web API (C# 12, Controller-based endpoints)                     |
| Architecture   | Microservices (Clean Architecture: Domain, Application, Infrastructure, API)   |
| Gateway        | YARP (Yet Another Reverse Proxy) on Port 5000 (Routing, CORS, Auth)            |
| Authentication | JWT (In-memory Access Token) + HttpOnly Secure SameSite Cookie Refresh         |
| Database       | SQLite (AuthService) + SQL Server Express LocalDB (Customer, Quote, Insurance) |
| Testing        | xUnit + Coverlet (80% Line/Branch Gate) & Vitest with jsdom (Angular)          |

# Microservices Topology & Data Ownership



# Request Pipeline & Clean Architecture

## 1. Angular 21 Client SPA

Functional interceptor attaches Bearer token, coalesces 401 refresh, injects correlation ID.

## 2. YARP API Gateway (:5000)

Validates JWT signature, echoes X-Correlation-ID, routes unchanged paths to target services.

## 3. BuildingBlocks Library

Shared error middleware, typed HTTP forwarders, pagination helpers, token extensions.

### ASP.NET CORE 8 • 4-PROJECT CLEAN ARCHITECTURE

#### API Layer (.API)

Controllers, routing definitions, model validation, Swagger docs.

#### Application Layer (.Application)

Commands, queries, DTOs, cross-service typed clients, domain exceptions.

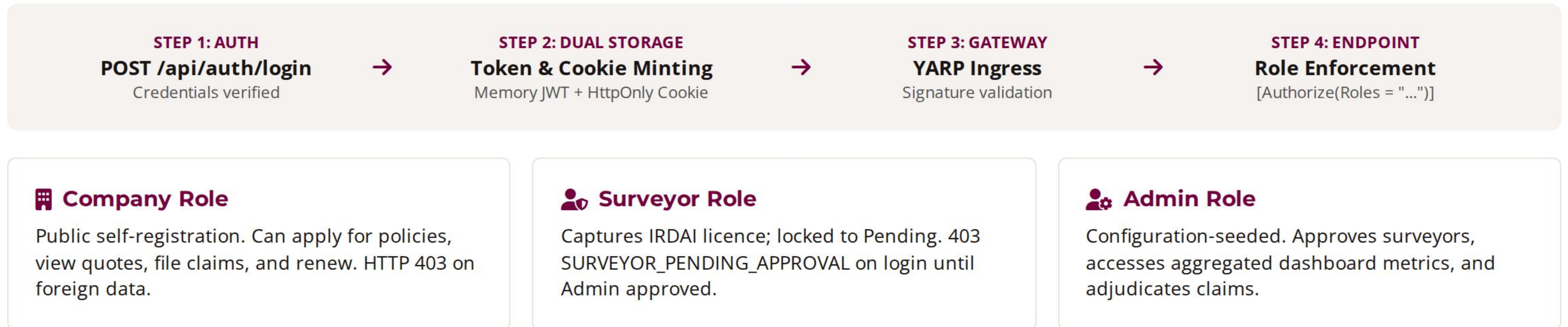
#### Domain Layer (.Domain)

Pure business logic: premium formulas, rules engine, period derivation.

#### Infrastructure Layer (.Infrastructure)

EF Core DbContexts, database migrations, Polly resilience handlers.

# JWT Authentication & Access Control



# Polyglot Persistence & Entity Ownership

## Auth Database (SQLite)

Lightweight relational store for identity and surveyor states.

### User Entity

Id, Email, PasswordHash, RefreshToken, Expiry

### SurveyorProfile Entity

Id, UserId (UQ), IrdaiLicenseNo, ValidUntil, Status

Standardized UTC DateTime for SQLite provider

## Core Business Stores (SQL Server LocalDB)

High-integrity relational databases for Customer, Quote, and Insurance.

### CustomerService

CompanyProfile (CIN, Contact, FleetSize, UserId)

### QuoteService

Quote (Estimates, Tier, AddOns, Status, ExpiryDate)

### InsuranceService Data Model

Policy (Term, Premium, RenewedFromId) • Claim (IncidentTime, DerivedPeriod, Decision) • TripEvent  
• FieldVerification • SurveyorReport

Standardized UTC DateTimeOffset for all SQL Server temporal logic

# System Reliability & Fault Tolerance



## Health Probes

Standardized /health endpoints exposed on YARP and all 4 microservices for liveness monitoring.



## Polly FX Fallback

3 retries with exponential backoff on live FX API ([open.er-api.com](https://open.er-api.com)). Exhaustion falls back to cached rate.



## Distributed Tracing

X-Correlation-ID header injected at Gateway and forwarded on all service calls via DelegatingHandler.



## Centralized Errors

Middleware maps domain exceptions (e.g., 404, 409, 410) deterministically to RFC-compliant JSON envelopes.

# Testing Strategy & 80% Coverage Gate

## Backend: xUnit + Coverlet

80% Line & Branch Gate

Targeted strictly to Application and Domain layers across all 4 services via table-driven theories:

- **Premium Formulas:** Fleet size × Base × Tier Multipliers + Add-ons
- **Rules Engine Matrix:** Tier × Derived Period × Claim Type × Add-ons
- **Renewal Boundaries:** Tested at 29%, 30%, 70%, and 100% loss ratios
- **Polly FX Retry:** Mocked handler fails twice → succeeds on 3rd; fallback verified
- **Atomic Claim Pickup:** Race-condition test against SQLite confirms single-winner 409

## Frontend: Vitest

jsdom DOM

Angular 21 CLI native integration testing core infrastructure:

- **Route Guards:** roleGuard, guestGuard, profile gate
- **HTTP Interceptor:** Coalesced 401 refresh & Bearer attach
- **Reactive Forms:** Chronological trip log validation
- **Session Services:** Initializer silent refresh checks

# Architectural Scope Decisions & Trade-offs

| ITEM  | DEFERRED CAPABILITY          | ARCHITECTURAL RATIONALE & CURRENT BASELINE                                                   |
|-------|------------------------------|----------------------------------------------------------------------------------------------|
| D3    | Standalone PaymentService    | Trainer-directed simplification; simulated deterministically in InsuranceService.            |
| D8    | Cloudinary / S3 File Uploads | Environment network constraint; streamlined to structured document repository link.          |
| D9    | Claims Resubmission Loop     | Replaced by immutable single-submission model; per-field audits preserved accurately.        |
| D10   | Underwritten Policy Binding  | Standardized product enables straight-through binding; >100% renewal loss surfaces as alert. |
| D1    | Password Reset Flow          | Deferred in favor of core JWT security, rotation cookies, and surveyor approval gates.       |
| D4/D7 | Advanced Multi-Field Sort    | Single-field server-side sorting implemented; multi-field cross-service sorting deferred.    |

# Thank You

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Open for Questions, Technical Discussion & Code Defense

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